

Hyeongjin Kim

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[GitHub: EliseEgkart](#) | [YouTube: Project demonstrations](#) | [Web Portfolio](#)

PROFILE

Researcher at the Control & Optimization Laboratory (COLAB), Hanyang University, pursuing an integrated M.S.–Ph.D. program in Electrical Engineering under the supervision of Prof. Jun Moon. My research interests focus on reinforcement learning, networks, robot learning, and robotic manipulation, supported by practical experience in ROS 2 robotics and sensor fusion.

RESEARCH INTERESTS

Reinforcement learning, networks and networked systems, robot learning, and robotic manipulation.

EDUCATION

- **Hanyang University** Sep. 2026 – Present
Integrated M.S.–Ph.D. program in Electrical Engineering Seoul, Republic of Korea
- **Incheon National University** Mar. 2021 – Aug. 2026
B.Eng. in Embedded Systems Engineering Incheon, Republic of Korea
 - Graduated first in class, Department of Embedded Systems Engineering
 - GPA: 4.28 / 4.5 (140 / 140 credits completed)

RESEARCH AND PROFESSIONAL EXPERIENCE

- **Control & Optimization Laboratory (COLAB)** Sep. 2026 – Present
Researcher (Advisor: Prof. Jun Moon) Seoul, Republic of Korea
 - Research interests: reinforcement learning, networks, robot learning, and robotic manipulation.
- **ICONS Laboratory** Mar. 2024 – Jun. 2026
Undergraduate Researcher (Advisor: Prof. Hyunbum Kim) Incheon, Republic of Korea
 - Investigated network optimization and resource efficiency in UAV-assisted ITS systems
 - Investigated IoT platform-based decision-making architectures in distributed system environments
- **Hyundai H-Mobility Class** Apr. 2025 – Jul. 2025
Autonomous Driving Track (Advanced Communication & Networking) Republic of Korea
 - Completed coursework in perception, decision-making, control, and communication.
 - ROS 2-based autonomous driving training / CAN, LIN, Ethernet, and V2X communication

PUBLICATIONS AND PRESENTATIONS

1. **H. Kim**, H. Kim, W. Kim, A. V. Vasilakos, and P. Bellavista, “Resting Drone-Enabled Enhanced ITS Coverage and V2X Integration Network Management for Urban Mobility Service,” *IEEE Transactions on Network and Service Management*, vol. 23, pp. 3268–3278, 2026. DOI: [10.1109/TNSM.2026.3673324](https://doi.org/10.1109/TNSM.2026.3673324).
2. **H. Kim**, J. Ben-Othman, B. Lee, and H. Kim, “Split Federated Learning-Enabled Deep Q-Networks for Generalized Path Planning in Distributed IoT Edge Platform,” *IEEE Internet of Things Journal*, vol. 13, no. 11, pp. 23495–23505, Jun. 2026. DOI: [10.1109/JIOT.2026.3670351](https://doi.org/10.1109/JIOT.2026.3670351).
3. **H. Kim**, H. Kim, and M. Guizani, “A Resting Drone-Enabled Surveillance Framework for Real-Time Monitoring With Resource Efficiency in IoT-Based Smart City,” *IEEE Internet of Things Journal*, vol. 12, no. 16, pp. 33995–34006, Aug. 2025. DOI: [10.1109/JIOT.2025.3577655](https://doi.org/10.1109/JIOT.2025.3577655).
4. **H. Kim** and H. Kim, “Analysis of Opportunistic Wi-Fi/LTE Connectivity Strategies for Outdoor Autonomous Driving Platforms in ITS Environments,” *KICS Fall Conference 2025*, Nov. 21, 2025. (Oral)
5. **H. Kim**, M. Kim, and H. Kim, “Analysis of V2V Shortest Path Search and Connection Strategies in ITS Environment Using Low-Power Short-Range Communication,” *KICS Fall Conference 2024*, Nov. 2024, pp. 1101–1102. (Poster)

PROJECTS (UNDERGRADUATE)

- **2026 RoboCup Smart Manufacturing : League Mobile Manipulator** *Jul. 2025 – Jul. 2026*
 - **Role:** Team Leader, Autonomous Driving Team
 - **Tech Stack:** ROS 2, Jetson Orin AGX, Rainbow Robotics RB-3, RH-P12-RN Gripper, Intel RealSense D435
 - **Key Features:**
 - * Autonomous navigation of a mobile manipulator to task-relevant workstations
 - * Vision-assisted pick-and-place manipulation using a 6-DoF robotic arm and parallel gripper
 - * Integrated perception, navigation, and manipulation pipelines for real-world task execution
- **Capstone Design : Last-Mile Delivery Robot** *Nov. 2025 – Jun. 2026*
 - **Role:** Team Leader
 - **Tech Stack:** ROS 2, 2D LiDAR (slamtec), Intel RealSense D435, IMU, Jetson Orin Nano, SLAM, Nav2, Moveit
 - **Key Features:**
 - * Enabled autonomous navigation to designated goal locations using localization results
 - * Estimated odometry using in-wheel motors with encoders and performed Cartographer-based SLAM
 - * Implemented vision-based elevator button recognition and IMU-based floor transition detection
- **FPV-Based RC Car Racing** *Nov. 2025 – Dec. 2025*
 - **Role:** Team Leader
 - **Tech Stack:** ESP32-C3, Arduino, BLE, ESP-NOW, FPV System (Camera & Goggles), Fusion 360
 - **Key Features:**
 - * Developed FPV-based RC car firmware supporting BLE and ESP-NOW communication
 - * Implemented real-time steering and speed control using joystick-based commands
 - * Designed and fabricated RC car chassis and bumper using Fusion 360 for racing durability
- **Real-World Autonomous Driving Vehicle (ERP-42 Platform)** *Apr. 2025 – Nov. 2025*
 - **Role:** Team Leader
 - **Tech Stack:** ROS 2, 3D LiDAR (VLP-16), Camera, RTK-GNSS, IMU, Vehicle Control
 - **Key Features:**
 - * Performed navigation driving using sensor fusion of RTK-GNSS and IMU
 - * Developed Pure Pursuit and PID-based vehicle control and onboard communication protocols
 - * Implemented object detection and obstacle avoidance logic using camera and Velodyne VLP-16 sensor fusion
- **Line-Following (CV) Autonomous RC Car** *May 2025 – Jun. 2025*
 - **Role:** Course Project
 - **Tech Stack:** Raspberry Pi, Arduino, OpenCV, Python, UART/Serial, PID Control
 - **Key Features:**
 - * Built a line-following autonomous RC car using camera-based computer vision for lane detection
 - * Integrated Raspberry Pi image processing with PID steering control and Arduino motor/servo actuation
 - * Implemented dual-mode control enabling manual RC and autonomous driving via UART communication
- **Virtual Gesture Interface Board** *Apr. 2025 – May 2025*
 - **Role:** Course Project
 - **Tech Stack:** JavaScript, p5.js, ml5.js (HandPose), Webcam, OBS
 - **Key Features:**
 - * Implemented hand gesture recognition using ml5.js HandPose
 - * Developed gesture-based mode switching and interactive on-screen reactions
 - * Integrated drawing palette interactions and streamed output via OBS Virtual Camera
- **Resting Drone** *Mar. 2024 – Nov. 2024*
 - **Role:** Team Leader
 - **Tech Stack:** Pixhawk 4, PX4 Autopilot, Optical Flow, LiDAR, Servo Motor Control, ECU & PM Board
 - **Key Features:**
 - * Developed an energy-efficient resting drone that attaches to structures to eliminate hovering
 - * Implemented servo-based attachment maneuvers with optical flow-based attitude stabilization
 - * Built a PX4-based autonomous flight control system using Pixhawk 4
- **Automated Tanghulu Manufacturing Machine** *Jan. 2024 – Apr. 2024*
 - **Role:** Entrepreneurship Club
 - **Tech Stack:** ESP32, OLED Display, DC Motor, Fusion 360
 - **Key Features:**
 - * Developed an OLED-based user interface for motor speed and operation time control
 - * Implemented joystick-based interface control and DC motor speed adjustment

* Contributed to filing four related patents (KR 10-2024-0040087 to 10-2024-0040090)

- **FPS Gun Console System**

Sep. 2023 – Dec. 2023

- **Role:** Team Leader
- **Tech Stack:** ESP32, Bluetooth Classic (SPP), Unity, OpenCV, IMU
- **Key Features:**
 - * Implemented a semi-virtualized IMU-based aiming system aligning muzzle orientation with on-screen view
 - * Enabled real-time input transmission via Bluetooth using ESP32 for PC FPS game integration
 - * Designed a Unity-based input emulation framework compatible with commercial FPS titles

- **Unity-Based Game Development Project**

Jul. 2023 – Sep. 2023

- **Role:** Personal Projects
- **Tech Stack:** Unity
- **Key Features:**
 - * Game Project 1: Bird Flight Simulation Game
 - * Game Project 2: 2D Side-Scrolling Action Platformer
 - * Game Project 3: Top-Down Arena Roguelike Action Game

- **Wireless Mobile Fan Control System : Fan-Li**

Sep. 2021 – Dec. 2021

- **Role:** Team Leader
- **Tech Stack:** Arduino, ESP32-CAM, Bluetooth Control App, Real-time Streaming, Fusion 360
- **Key Features:**
 - * Enabled wireless control of the fan via a Bluetooth-based mobile app
 - * Implemented real-time video streaming using ESP32-CAM
 - * Integrated mobility and airflow control into a mobile fan system

- **Intelligent Book Arrangement Robot (IBA)**

May 2021 – Sep. 2021

- **Role:** Team Member
- **Tech Stack:** Arduino, RFID, DC Motor, Servo Motor, Stepper Motor
- **Key Features:**
 - * Implemented RFID-based localization using floor-mounted tags
 - * Built a multi-motor control system for navigation and book handling
 - * Supported vision-based book classification for automated sorting

ADDITIONAL ROBOTICS PROJECT

- **Camera–LiDAR Sensor Fusion Smart Stroller**

ROS 2, camera, 2D LiDAR, sensor fusion, Arduino

- Integrated camera and 2D LiDAR sensing through a ROS 2 sensor-fusion system, with low-level motor actuation controlled by Arduino.

LEADERSHIP & AMBASSADOR ACTIVITIES

- **Autonomous Driving Track Ambassador, Hyundai H-Mobility Class**

Apr. 2025 – Jul. 2025

Hyundai Motor Group

- Selected as an Autonomous Driving Track ambassador and produced weekly educational summary content and promotional materials for the program.

- **President, UNITA Robotics & Mobility Club (3rd Cohort)**

Jan. 2026 – Present

Incheon National University

- Designed, lectured, and led a ROS 2-based autonomous robotics curriculum, incorporating quizzes, student presentations, and hands-on projects.

- **President, UNITA Robotics & Mobility Club (2nd Cohort)**

Apr. 2025 – Dec. 2025

Incheon National University

- Led an autonomous mobility competition team and developed a sensor fusion-based autonomous driving system using 3D LiDAR, cameras, GNSS-RTK and IMU

TECHNICAL SKILLS (CORE TOOLS I USE DAILY)

ROS 2 | C/C++ | Python | Arduino (Platform) | Linux (Ubuntu) | Git/GitHub | Markdown | LaTeX

AWARDS

- **RoboCup Industrial, Smart Manufacturing League**

Jul. 2026

2026 World Champion, First Place

Role: Team Leading Member

- **Outstanding Learner Award, Hyundai H-Mobility Class**
Hyundai Motor Group R&D Division Head Award

Aug. 25, 2025
- **Grand Prize, Department of Embedded Systems Engineering Graduation Project Competition**
College of Information Technology Dean's Award, Incheon National University
 Role: Team leader

Jun. 2026
- **Grand Prize, Self-Built Electric Vehicle Competition**
Future Automotive Convergence Innovation Center Director Award, Incheon National University
 Role: Team leader

Dec. 2025
- **Special Prize, National Creative Capstone Design Competition**
Korea Technopark Promotion Association President Award
 Role: Team leader

Nov. 2024
- **Bronze Prize, Sungkyunkwan University Consortium Creative Capstone Design Competition**
Engineering Education Innovation Center Director Award, Sungkyunkwan University
 Role: Team leader

Nov. 2024
- **Grand Prize, Incheon National University Creative Capstone Design Competition**
Incheon National University President Award
 Role: Team leader

Sep. 2024
- **Grand Prize, Software Contest of the College of Information Technology**
College of Information Technology Dean's Award, Incheon National University
 Role: Team leader

Dec. 2023
- **Grand Prize, INU Maker Competition**
Incheon National University President Award
 Role: Team leader

Jan. 2022
- **Honorable Mention, Smart Mobility Competition**
Engineering Education Innovation Center Director Award, Incheon National University
 Role: Team leader

Dec. 2021
- **Bronze Prize, INU Autonomous Driving Software Competition**
College of Information Technology Dean's Award, Incheon National University
 Role: Team member

Nov. 2021

ACADEMIC SERVICE

- **Teaching Assistant, Matrix and Algorithms**
Department of Embedded Systems Engineering, Incheon National University
 Matrix: Fall 2025, Spring 2026; Algorithms: Fall 2025

Fall 2025; Spring 2026

ACADEMIC DISTINCTIONS AND OFFICIAL RECOGNITION

Incheon National University, Aug. 21, 2026:

- President's Recommendation for Outstanding Talent.
- Dean's Commendation for Academic Excellence and Exemplary Conduct.
- INU SURPRISE Outstanding Talent Certificate.